NTNU
Kunnskap for en bedre verden

Department of Engineering Cybernetics

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Comparison of Skyfield SGP4 and Basilisk Numerical Orbit Propagation for GNSS-R Formation-Flying Applications

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CHAPTER

ONE

INTRODUCTION

https://www.itk.ntnu.no/emner/masteroppgave

This has to be included in the report to make it clear to the sensors what has been done:

- Information, software and equipment available/used
- What help has been received during the work
- The use of AI. The AI-declaration schema MUST be inserted on the first page after the title page

1.0.1 FNs sustainability goals

1.0.2 Background and motivation

1.0.3 Research problem

1.0.4 Research objectives

1.0.5 Research questions

CHAPTER

TWO

LITERATURE REVIEW

2.0.1 Orbit propagation methods

- SGP4 limitations - Numerical propagation - High-fidelity orbit dynamics

2.0.2 Atmospheric orbital perturbation

- NRLMSISE-00 - Space weather effects - Attitude dependent drag

2.0.3 Formation flying

- Formation maintenance - Station keeping - GNSS-R formation types - GNSS-R formation sensitivity

2.0.4 Spacecraft Lifetime and Power Modeling

- CubeSat power systems - Eclipse modeling - Lifetime estimation methods

2.0.5 CubeSat components

CHAPTER

THREE

BACKGROUND THEORY

3.1 Reference frames

3.1.1 ECI

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CHAPTER

FOUR

METHODOLOGY

CHAPTER

FIVE

RESULTS

Plotting ideas for part 2:

- The big important plot: Fuel level over time
- Each subsystem should be validated in addition to showing the full fuel consumption over simulation duration:
  - RW speed/torque step response
  - Thruster force output as a function of fuel level (/tank pressure)
  - MRP feedback controller tracking error and/or step response (would also be interesting to see a step response for different fuel levels)
  - Decomposed net power and battery level (should also show the illumination factor in the same/adjacent plot and the FSW state)

CHAPTER

SIX

DISCUSSION

CHAPTER

SEVEN

CONCLUSIONS

7.1 Conclusion

REFERENCES

[1] Weichao Zhong and Pini Gurfil. "Mean Orbital Elements Estimation for Autonomous Satellite Guidance and Orbit Control". In: *Journal of Guidance, Control, and Dynamics* 36.6 (2013), pp. 1624–1641. DOI: 10.2514/1.60701. URL: https://www.researchgate.net/publication/260501946_Mean_Orbital_Elements_Estimation_for_Autonomous_Satellite_Guidance_and_Orbit_Control.

APPENDICES

A - GITHUB REPOSITORY

All simulator code and LaTeX-files used in this document are included in the GitHub repositories linked below.

Specialization project report LaTeX repository link

- https://github.com/chystad/Specialization-Project-Report-LaTeX-

Orbit Simulator repository link

- https://github.com/chystad/Orbit_Simulator

B - NUMERICAL VALUES

Numerical Values for Physical Constants

$Constant$	$Value$	$Unit$
J_0	-1	-
J_1	0	-
J_2	1.08263×10^{-3}	-
J_3	-2.53266×10^{-6}	-
J_4	-1.61962×10^{-6}	-
P_{Sun}	$4.578299965 \times 10^{-6}$	Ws/m^3
$1AU$	149597871	km
R	6378136.6	m
H_0	7200	m
ρ_0	1.225	kg/m^3
$-$	$-$	$-$
$-$	$-$	$-$
$-$	$-$	$-$

Table .0.1: Physical constants and associated numerical value. J-terms courtesy of [1].